# Checkpoint 1 — Write short answers:

1. How many TCP, TCP-port-80, and UDP-port-53 packets did you find?
   TCP packets: 474035
   TCP port 80 packets: 22430
   UDP port 53 packets: 57
2. Why might a filter return zero packets?
   A filter might return zero packets if no packets in the pcap file match the specified protocol or port. If filtering for TCP port 80 returns zero packets, it means the capture contains no TCP traffic using port 80, or other reasons like a wrong port number, wrong protocol filtering or an invalid filter condition
3. What is the difference between filtering by protocol and filtering by port?
   Filtering by protocol selects packets based on the protocol they use, like TCP or UDP. TCP in pkt selects packets containing a TCP layer regardless of the port number. Filtering by port selects packets based on their source or destination port numbers. For example, TCP port 80 selects traffic from port 80, which is commonly known for HTTP traffic. Protocol filtering identifies the type of network traffic; port filtering narrows down the result to traffic related to specific services.

```
python3 part1.py
Number of packets: 495056
Ether / ARP who has 147.32.84.165 says 147.32.84.1 / Padding
Ether / ARP who has 147.32.84.165 says 147.32.84.1 / Padding
Ether / ARP is at 08:00:27:b5:b7:19 says 147.32.84.165 / Padding
Ether / ARP is at 08:00:27:b5:b7:19 says 147.32.84.165 / Padding
Ether / ARP who has 147.32.84.165 says 147.32.84.1 / Padding
Ether / ARP is at 08:00:27:b5:b7:19 says 147.32.84.165 / Padding
Is this TCP? False
TCP packets: 474035
Ether / IP / TCP 78.40.125.4:ircd > 147.32.84.165:1181 PA / Raw
Ether / IP / TCP 147.32.84.165:1181 > 78.40.125.4:ircd PA / Raw
Ether / IP / TCP 147.32.84.165:1181 > 78.40.125.4:ircd PA / Raw
Ether / IP / TCP 78.40.125.4:ircd > 147.32.84.165:1181 A / Padding
Ether / IP / TCP 78.40.125.4:ircd > 147.32.84.165:1181 PA / Raw
Ether / IP / TCP 147.32.84.165:1181 > 78.40.125.4:ircd A / Padding
Ether / IP / TCP 147.32.84.165:1181 > 78.40.125.4:ircd A / Padding
Ether / IP / TCP 78.40.125.4:ircd > 147.32.84.165:1181 PA / Raw
Ether / IP / TCP 147.32.84.165:1181 > 78.40.125.4:ircd PA / Raw
Ether / IP / TCP 147.32.84.165:1181 > 78.40.125.4:ircd PA / Raw
Ether / IP / TCP 78.40.125.4:ircd > 147.32.84.165:1181 A / Padding
Ether / IP / TCP 78.40.125.4:ircd > 147.32.84.165:1181 PA / Raw
Ether / IP / TCP 147.32.84.165:1181 > 78.40.125.4:ircd A / Padding
Ether / IP / TCP 147.32.84.165:1181 > 78.40.125.4:ircd A / Padding
Ether / IP / TCP 147.32.84.165:1181 > 78.40.125.4:ircd PA / Raw
Ether / IP / TCP 147.32.84.165:1181 > 78.40.125.4:ircd PA / Raw
Ether / IP / TCP 78.40.125.4:ircd > 147.32.84.165:1181 A / Padding
Ether / IP / TCP 78.40.125.4:ircd > 147.32.84.165:1181 PA / Raw
```

Ether / IP / TCP 147.32.84.165:1181 > 78.40.125.4:ircd PA / Raw
Ether / IP / TCP 147.32.84.165:1181 > 78.40.125.4:ircd PA / Raw
Ether / IP / TCP 78.40.125.4:ircd > 147.32.84.165:1181 A / Padding
Ether / IP / TCP 78.40.125.4:ircd > 147.32.84.165:1181 PA / Raw
Ether / IP / TCP 147.32.84.165:1181 > 78.40.125.4:ircd PA / Raw
Ether / IP / TCP 147.32.84.165:1181 > 78.40.125.4:ircd PA / Raw
Ether / IP / TCP 78.40.125.4:ircd > 147.32.84.165:1181 A / Padding
Ether / IP / TCP 78.40.125.4:ircd > 147.32.84.165:1181 PA / Raw
Ether / IP / TCP 147.32.84.165:1181 > 78.40.125.4:ircd PA / Raw
Ether / IP / TCP 147.32.84.165:1181 > 78.40.125.4:ircd PA / Raw
Ether / IP / TCP 78.40.125.4:ircd > 147.32.84.165:1181 A / Padding
Ether / IP / TCP 78.40.125.4:ircd > 147.32.84.165:1181 PA / Raw
Ether / IP / TCP 147.32.84.165:1181 > 78.40.125.4:ircd PA / Raw
Ether / IP / TCP 147.32.84.165:1181 > 78.40.125.4:ircd PA / Raw
Ether / IP / TCP 78.40.125.4:ircd > 147.32.84.165:1181 A / Padding
Ether / IP / TCP 78.40.125.4:ircd > 147.32.84.165:1181 PA / Raw
Ether / IP / TCP 147.32.84.165:1181 > 78.40.125.4:ircd PA / Raw
Ether / IP / TCP 147.32.84.165:1181 > 78.40.125.4:ircd PA / Raw
Ether / IP / TCP 78.40.125.4:ircd > 147.32.84.165:1181 A / Padding
Ether / IP / TCP 147.32.84.165:1181 > 78.40.125.4:ircd PA / Raw
Ether / IP / TCP 147.32.84.165:1181 > 78.40.125.4:ircd PA / Raw
Ether / IP / TCP 78.40.125.4:ircd > 147.32.84.165:1181 A / Padding
Ether / IP / TCP 147.32.84.165:1181 > 78.40.125.4:ircd PA / Raw
Ether / IP / TCP 147.32.84.165:1181 > 78.40.125.4:ircd PA / Raw
Ether / IP / TCP 78.40.125.4:ircd > 147.32.84.165:1181 A / Padding
Ether / IP / TCP 147.32.84.165:1181 > 78.40.125.4:ircd PA / Raw
Ether / IP / TCP 147.32.84.165:1181 > 78.40.125.4:ircd PA / Raw
Ether / IP / TCP 78.40.125.4:ircd > 147.32.84.165:1181 A / Padding
Ether / IP / TCP 78.40.125.4:ircd > 147.32.84.165:1181 PA / Raw
Ether / IP / TCP 147.32.84.165:1181 > 78.40.125.4:ircd PA / Raw
Ether / IP / TCP 147.32.84.165:1181 > 78.40.125.4:ircd PA / Raw
Ether / IP / TCP 78.40.125.4:ircd > 147.32.84.165:1181 A / Padding
Source port: 6667
Destination port: 1181
TCP port 80 packets: 22430
Ether / IP / TCP 147.32.84.165:1183 > 111.89.136.28:http S
Ether / IP / TCP 147.32.84.165:1183 > 111.89.136.28:http S
Ether / IP / TCP 147.32.84.165:1184 > 111.89.136.29:http S
Ether / IP / TCP 147.32.84.165:1184 > 111.89.136.29:http S
Ether / IP / TCP 147.32.84.165:1185 > 111.89.136.30:http S
Ether / IP / TCP 147.32.84.165:1185 > 111.89.136.30:http S
Ether / IP / TCP 147.32.84.165:1186 > 111.89.136.31:http S
Ether / IP / TCP 147.32.84.165:1186 > 111.89.136.31:http S
Ether / IP / TCP 147.32.84.165:1187 > 111.89.136.32:http S
Ether / IP / TCP 147.32.84.165:1187 > 111.89.136.32:http S
Ether / IP / TCP 147.32.84.165:1188 > 111.89.136.33:http S
Ether / IP / TCP 147.32.84.165:1188 > 111.89.136.33:http S
Ether / IP / TCP 147.32.84.165:1189 > 111.89.136.34:http S
Ether / IP / TCP 147.32.84.165:1189 > 111.89.136.34:http S
Ether / IP / TCP 147.32.84.165:1190 > 111.89.136.35:http S

Ether / IP / TCP 147.32.84.165:1190 > 111.89.136.35:http S
Ether / IP / TCP 147.32.84.165:1191 > 111.89.136.36:http S
Ether / IP / TCP 147.32.84.165:1191 > 111.89.136.36:http S
Ether / IP / TCP 147.32.84.165:1192 > 111.89.136.37:http S
Ether / IP / TCP 147.32.84.165:1192 > 111.89.136.37:http S
Ether / IP / TCP 111.89.136.28:http > 147.32.84.165:1183 SA
Ether / IP / TCP 147.32.84.165:1193 > 111.89.136.38:http S
Ether / IP / TCP 147.32.84.165:1193 > 111.89.136.38:http S
Ether / IP / TCP 147.32.84.165:1183 > 111.89.136.28:http A / Padding
Ether / IP / TCP 147.32.84.165:1183 > 111.89.136.28:http A / Padding
Ether / IP / TCP 147.32.84.165:1183 > 111.89.136.28:http FA / Padding
Ether / IP / TCP 147.32.84.165:1183 > 111.89.136.28:http FA / Padding
Ether / IP / TCP 111.89.136.29:http > 147.32.84.165:1184 SA
Ether / IP / TCP 147.32.84.165:openvpn > 111.89.136.39:http S
Ether / IP / TCP 147.32.84.165:openvpn > 111.89.136.39:http S
Ether / IP / TCP 147.32.84.165:1184 > 111.89.136.29:http A / Padding
Ether / IP / TCP 147.32.84.165:1184 > 111.89.136.29:http A / Padding
Ether / IP / TCP 147.32.84.165:1184 > 111.89.136.29:http FA / Padding
Ether / IP / TCP 147.32.84.165:1184 > 111.89.136.29:http FA / Padding
Ether / IP / TCP 111.89.136.31:http > 147.32.84.165:1186 SA
Ether / IP / TCP 147.32.84.165:1195 > 111.89.136.40:http S
Ether / IP / TCP 147.32.84.165:1195 > 111.89.136.40:http S
Ether / IP / TCP 147.32.84.165:1186 > 111.89.136.31:http A / Padding
Ether / IP / TCP 147.32.84.165:1186 > 111.89.136.31:http A / Padding
Ether / IP / TCP 147.32.84.165:1186 > 111.89.136.31:http FA / Padding
Ether / IP / TCP 147.32.84.165:1186 > 111.89.136.31:http FA / Padding
Ether / IP / TCP 111.89.136.30:http > 147.32.84.165:1185 SA
Ether / IP / TCP 147.32.84.165:1196 > 111.89.136.41:http S
Ether / IP / TCP 147.32.84.165:1196 > 111.89.136.41:http S
Ether / IP / TCP 147.32.84.165:1185 > 111.89.136.30:http A / Padding
Ether / IP / TCP 147.32.84.165:1185 > 111.89.136.30:http A / Padding
Ether / IP / TCP 147.32.84.165:1185 > 111.89.136.30:http FA / Padding
Ether / IP / TCP 147.32.84.165:1185 > 111.89.136.30:http FA / Padding
Ether / IP / TCP 111.89.136.33:http > 147.32.84.165:1188 SA
Ether / IP / TCP 111.89.136.32:http > 147.32.84.165:1187 SA
UDP port 53 packets: 57
Ether / IP / UDP / DNS Qry b'irc.freenode.net.'
Ether / IP / UDP / DNS Qry b'irc.freenode.net.'
Ether / IP / UDP / DNS Qry b'irc.freenode.net.'
Ether / IP / UDP / DNS Ans b'chat.freenode.net.'
Ether / IP / UDP / DNS Qry b'irc.freenode.net.'
Ether / IP / UDP / DNS Ans b'chat.freenode.net.'
Ether / IP / UDP / DNS Qry b'wpad.'
Ether / IP / UDP / DNS Qry b'wpad.'
Ether / IP / UDP / DNS Ans name-error
Ether / IP / UDP / DNS Qry b'cr-tools.clients.google.com.'
Ether / IP / UDP / DNS Qry b'cr-tools.clients.google.com.'
Ether / IP / UDP / DNS Ans b'clients.l.google.com.'
Ether / IP / UDP / DNS Qry b'irc.freenode.net.'
Ether / IP / UDP / DNS Qry b'irc.freenode.net.'

```
Ether / IP / UDP / DNS Ans b'chat.freenode.net.'
Ether / IP / UDP / DNS Qry b'javadl-esd.sun.com.'
Ether / IP / UDP / DNS Qry b'javadl-esd.sun.com.'
Ether / IP / UDP / DNS Ans b'javadl-esd-sino.sun.com.edgesuite.net.'
Ether / IP / UDP / DNS Qry b'irc.freenode.net.'
Ether / IP / UDP / DNS Qry b'irc.freenode.net.'
Ether / IP / UDP / DNS Qry b'wpad.'
Ether / IP / UDP / DNS Qry b'wpad.'
Ether / IP / UDP / DNS Ans name-error
Ether / IP / UDP / DNS Ans b'chat.freenode.net.'
Ether / IP / UDP / DNS Qry b'cr-tools.clients.google.com.'
Ether / IP / UDP / DNS Qry b'cr-tools.clients.google.com.'
Ether / IP / UDP / DNS Ans b'clients.l.google.com.'
Ether / IP / UDP / DNS Qry b'irc.freenode.net.'
Ether / IP / UDP / DNS Qry b'irc.freenode.net.'
Ether / IP / UDP / DNS Qry b'wpad.'
Ether / IP / UDP / DNS Qry b'wpad.'
Ether / IP / UDP / DNS Ans name-error
Ether / IP / UDP / DNS Ans b'chat.freenode.net.'
Ether / IP / UDP / DNS Qry b'cr-tools.clients.google.com.'
Ether / IP / UDP / DNS Qry b'cr-tools.clients.google.com.'
Ether / IP / UDP / DNS Ans b'clients.l.google.com.'
Ether / IP / UDP / DNS Qry b'dl.javafx.com.'
Ether / IP / UDP / DNS Qry b'dl.javafx.com.'
Ether / IP / UDP / DNS Ans 137.254.16.78
Ether / IP / UDP / DNS Qry b'www.google.com.'
Ether / IP / UDP / DNS Qry b'www.google.com.'
Ether / IP / UDP / DNS Ans b'www.l.google.com.'
Ether / IP / UDP / DNS Qry b'www.nmap.org.'
Ether / IP / UDP / DNS Qry b'www.nmap.org.'
Ether / IP / UDP / DNS Ans 74.207.254.18
Ether / IP / UDP / DNS Qry b'nmap.org.'
Ether / IP / UDP / DNS Qry b'nmap.org.'
Ether / IP / UDP / DNS Ans 74.207.254.18
Ether / IP / UDP / DNS Qry b'wpad.'
Ether / IP / UDP / DNS Qry b'wpad.'
```

# Checkpoint 2 — Answer:

1. What do source and destination IP addresses tell you?
   The source IP address identifies the device that sent the packet, while the destination IP address identifies the device intended to receive it. These addresses help investigators identify communication hosts and trace traffic between potentially compromised devices and command and control servers.
2. What is the purpose of TCP flags?
   TCP flags indicate the state and purpose of a TCP connection, SYN initiates a connection, ACK acknowledges the data being received, FIN indicates that communication is finished and RST resets a

connection. Investigators can examine these flags to identify unusual connection patterns, such as repeated SYN packets that may indicated a SYN flood attack.

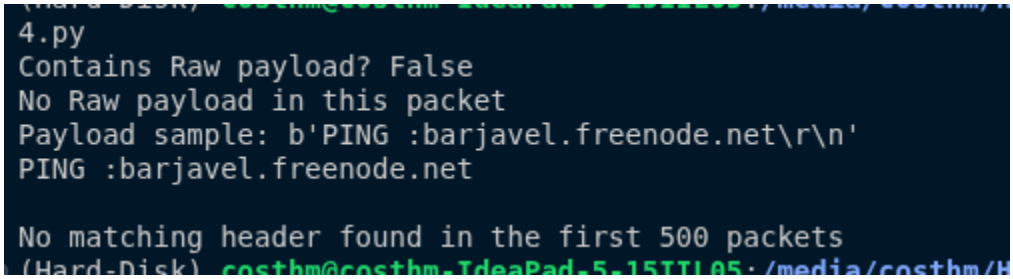
3. Why are timestamps useful when investigating possible flooding
   Timestamps show when packets were captured, allowing investigators to examine the frequency and timing of network traffic. A large number of packets arriving within a short period may indicate a flood attack. Timestamp also help identify traffic spikes, establish event sequences, and relates with network activity with other security logs.

```
python3 part2.py
###[ Ethernet ]###
  dst       = 08:00:27:b5:b7:19
  src       = 00:1e:49:db:19:c3
  type      = ARP
###[ ARP ]###
     hwtype    = Ethernet (10Mb)
     ptype     = IPv4
     hwlen     = 6
     plen      = 4
     op        = who-has
     hwsrc     = 00:1e:49:db:19:c3
     psrc      = 147.32.84.1
     hwdst     = 08:00:27:b5:b7:19
     pdst      = 147.32.84.165
###[ Padding ]###
        load      = b'\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00\x00'

Contains IPv4? False
78.40.125.4 -> 147.32.84.165
147.32.84.165 -> 78.40.125.4
147.32.84.165 -> 78.40.125.4
78.40.125.4 -> 147.32.84.165
78.40.125.4 -> 147.32.84.165
Size (bytes): 60
Timestamp: 1313153171.504306
Source: 78.40.125.4 Destination: 147.32.84.165 Bytes: 83 Time: 1313153244.357711
Source: 147.32.84.165 Destination: 78.40.125.4 Bytes: 83 Time: 1313153244.358073
Source: 147.32.84.165 Destination: 78.40.125.4 Bytes: 83 Time: 1313153244.358085
Source: 78.40.125.4 Destination: 147.32.84.165 Bytes: 60 Time: 1313153244.388656
Source: 78.40.125.4 Destination: 147.32.84.165 Bytes: 120 Time: 1313153273.62872
```

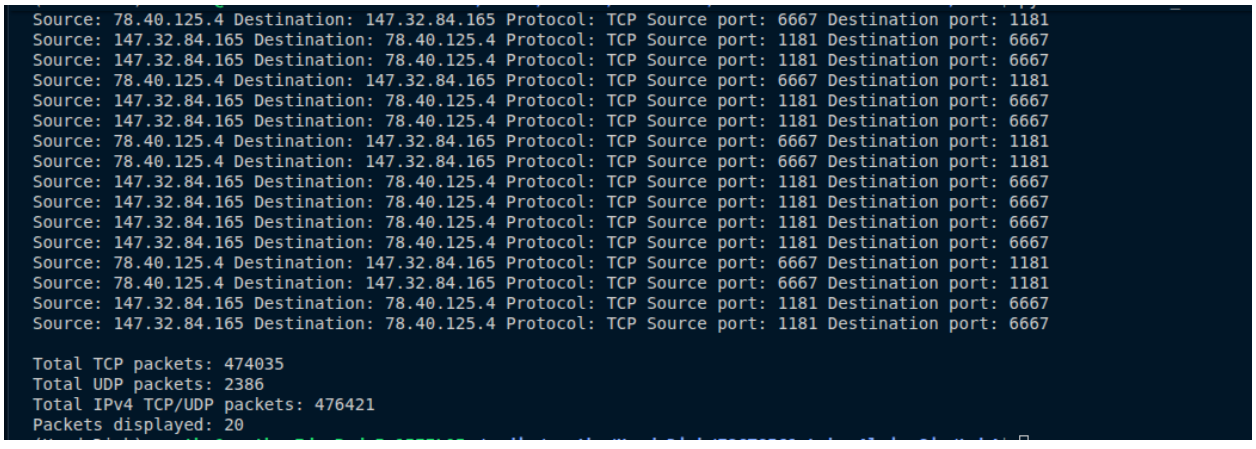
# Checkpoint 3 — Answer:

1. Did you find readable payload text




   Yes. The packet capture contains readable payload like “PING” and a plaintext URL, which can suggest some network communication was transmitted in plaintext and could be inspected directly from the packet capture
2. What might plaintext HTTP expose?
   Plaintext HTTP traffic may expose requested URLs, HTTP headers, browser information such as the User-Agent, form submissions, or user passwords if transmitted without encryption. This information could help an attacker understand their victim’s activity and identify network communication weak points
3. Why can you not inspect the contents of encrypted HTTPS traffic using these simple commands?
   HTTPS encrypts application data using TLS, so the payload is not readable in plaintext in a packet capture. Scapy can still display packet headers nad the encrypted payload, but not directly exposing the original HTTP content without the decryption key and TLS decryption procedures. This encryption protects sensitive information like passwords or session cookies from being easily read by someone capturing network traffic.

# Part 4



# Part 5

```
(Hard-Disk) costhm@costhm-IdeaPad-5-15IIL05:/media/costhm/Hard-Disk/FSCT8561-Labs-Alvin-Sin/Lab4$ python3 anomaly_detector.py
ALERT: Unusually frequent traffic from 147.32.84.165 - 21 packets within 5 seconds
ALERT: Unusually frequent traffic from 78.40.125.4 - 21 packets within 5 seconds
ALERT: Unusually frequent traffic from 85.190.0.3 - 21 packets within 5 seconds
ALERT: Unusually frequent traffic from 208.86.166.68 - 21 packets within 5 seconds
ALERT: Unusually frequent traffic from 64.31.13.148 - 21 packets within 5 seconds
ALERT: Unusually frequent traffic from 178.77.71.27 - 21 packets within 5 seconds
ALERT: Unusually frequent traffic from 109.74.55.27 - 21 packets within 5 seconds

--- Detection Summary ---
Total TCP packets: 474035
Total UDP packets: 2386
Total IPv4 TCP/UDP packets: 476421
Number of suspicious IPs: 7
```

# Part 6 — Reflection questions

1. **Why is packet inspection important for network security?**
   Packet inspection improves security analysts in identifying suspicious network activities, investigate communication between devices, detect unusual traffic patterns, and identify potential attacks
2. **What types of security risks can arise from unencrypted traffic?**
   Unencrypted traffic can expose sensitive informations, An attacker who captured the traffic can steal credentials, impersonate users or monitor communications
3. **What are the limitations of passive packet sniffing and PCAP analysis?**
   Passive sniffing only captures traffic visible to the monitoring point, so some communications may be missed. Encrypted payloads are difficult to inspect without decryption keys, and PCAP analysis may not provide enough context to determine if a communication is malicious.
4. **How does encryption help prevent information leakage?**
   Encryption turns readable data into ciphertext that cannot be read in plaintext without decryption keys, helps with protecting sensitive information from being read by unauthorized parties who capture network traffic.
5. **Why might a simple packet-rate detector produce false positives?**
   Legit devices and service can generate high volume of traffic in a short time, triggering a threshold that was set. A detector can only measure packet frequency but not able to understand and separate legit traffic from malicious traffic

# Part 7

The analysis of the provided botnet-capture-20110812-rbot.pcap file demonstrated how packet inspection can be used to examine network communications and identify potential security risks. Using Scapy, I inspected packet protocols, source and destination IP addresses, port numbers, TCP flags, timestamps, and payload data. One readable payload contained the text PING :barjavel.freenode.net, which appears to be an IRC protocol message. This observation suggests that the capture includes IRC-related communication, although this packet alone does not prove malicious activity. I also examined how TCP port 80 traffic could be identified and how packet timestamps can help reveal unusual traffic patterns. The first packet inspected did not contain a Raw payload, demonstrating that not every packet contains directly accessible application data.

One potential security risk is the use of unencrypted protocols. If sensitive information is transmitted in plaintext, an attacker who can capture the traffic may be able to read communications, identify system information, or obtain credentials. IRC communication may also be relevant when investigating botnet activity because some malware uses IRC servers for command-and-control communication. However, confirming this would require further evidence, such as examining communication patterns, payload contents, and connections to known malicious infrastructure.

To reduce exposure, organizations should use encrypted protocols such as HTTPS and SSH instead of their unencrypted alternatives, restrict unnecessary network services, and monitor outbound connections to suspicious destinations. Security analysts should investigate unusual connection patterns, repeated requests, unexpected ports, and abnormal packet rates. A packet-rate detector can help identify a source IP generating more than 20 IPv4 TCP or UDP packets within five seconds. However, an alert should trigger further investigation rather than be treated as proof of an attack.

There are also limitations to this analysis. Passive packet capture only records traffic visible at the monitoring point, and encrypted HTTPS payloads cannot normally be read without appropriate decryption capabilities. A PCAP may lack the surrounding system logs or endpoint evidence needed to determine the purpose of a connection. Additionally, a simple threshold-based detector can produce false positives when legitimate services generate bursts of traffic. It can also miss distributed attacks when traffic is spread across many source IP addresses. Therefore, packet inspection should be combined with other security monitoring methods and contextual evidence to support reliable conclusions.